Shahjalal University of Science and Technology

Department of Electrical and Electronic Engineering

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Experiment 05

Verification of Superposition Theorem for a DC Network

Submitted By:
Raj Bhowmik Sany
Reg. No: 202561901048
Session: 2025
Department of Software Engineering

Submitted To:
Dr. Md Rasedujjaman
Associate Professor
Department of Electrical and Electronic Engineering
Shahjalal University of Science and Technology

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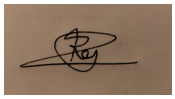
Declaration

I hereby declare that the work presented in this laboratory report entitled **"Verification of Superposition Theorem for a DC Network"** is an authentic record of my own work carried out for the course **Basic Electrical and Electronic Circuit Lab (EEE 0713 1102W)**.

This report has been prepared by me independently and has not been submitted elsewhere for the award of any degree, diploma, or credit.

Sylhet, Bangladesh

Date: August 31, 2026



Raj Bhowmik Sany
Reg. No: 202561901048

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Objective

- To understand the principle of the Superposition Theorem and its application in linear, bilateral DC resistive networks containing more than one independent source.
- To experimentally verify that the total branch current in a multi-source linear circuit equals the algebraic sum of the individual branch currents produced by each independent source acting alone.
- To compare the measured branch currents against theoretical calculations and evaluate the percentage error.

Introduction

Many practical DC circuits are driven by more than one independent source at once, and analyzing them directly can get complicated fast. The **Superposition Theorem** gives a way around that: instead of solving the whole multi-source circuit in one go, it lets us analyze the effect of each source on its own – with all the other independent sources turned off – and then add the individual results back together to get the total response. This experiment builds a two-source resistive network, measures a branch current with both sources active and with each source active alone, and checks that the superposition relationship holds.

Theory

Statement of the Theorem

> *In any linear, bilateral electrical network containing more than one independent source, the current through or voltage across any branch is equal to the algebraic sum of the currents or voltages produced by each independent source acting individually, while all other independent sources are deactivated (replaced by their internal impedances).*

Deactivation of Independent Sources

- **Ideal Voltage Sources** are deactivated by setting their voltage to zero, i.e. replacing them with a **short circuit**.
- **Ideal Current Sources** are deactivated by setting their current to zero, i.e. replacing them with an **open circuit**.

Mathematical Formulation

For a branch carrying current I in a circuit driven by two independent voltage sources V_1 and V_2,

$$I = I' + I''$$

where I' is the branch current when only V_1 is active (V_2 short-circuited), and I'' is the branch current when only V_2 is active (V_1 short-circuited).

Non-Applicability to Power Calculations

One thing to be careful about is that superposition does **not** work for power. Power depends on current (or voltage) squared, $P = I^2R$, and squaring is a non-linear operation, so you can't just add up the individual powers from each source:

$$P_{\text{total}} \neq P_1 + P_2.$$

The right way to do it is to find the total current or voltage first using superposition, and only then plug that into the power formula.

Apparatus and Components

Table 1: List of Equipment and Components Used

Item Description	Specification / Rating	Quantity
Dual-Channel DC Power Supply	RPS1, RPS2 (0 V–30 V)	1
Resistor R_1	325 Ω	1
Resistor R_2	550 Ω	1
Resistor R_3	960 Ω	1
Digital Multimeter (DMM)	DC Volts / DC Amps	1
Solderless Breadboard	Standard Tie-Point	1
Connecting Wires	Jumper Wires	As required

Circuit Diagrams

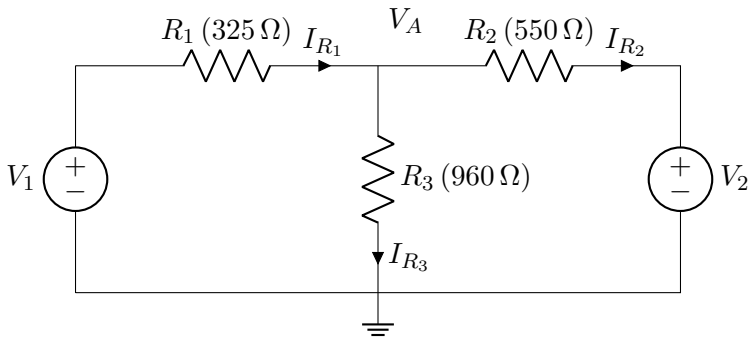


Figure 1: Complete DC network with both independent voltage sources V_1 and V_2 active. The branch current I_{R_3} is the response under test.

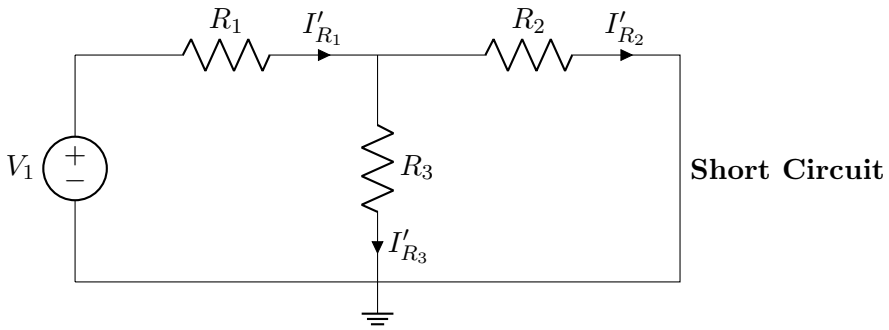


Figure 2: Modified circuit with V_1 active and V_2 replaced by a short circuit, giving the isolated contribution I'_{R_3}.

Experimental Procedure

1. Build the circuit shown in Figure 1 on the breadboard, using $R_1 = 325\,\Omega$, $R_2 = 550\,\Omega$, and $R_3 = 960\,\Omega$.

2. Turn on both RPS1 (V_1) and RPS2 (V_2), set them to the first trial voltage combination, and measure the total branch current I_{R_3} flowing through R_3.

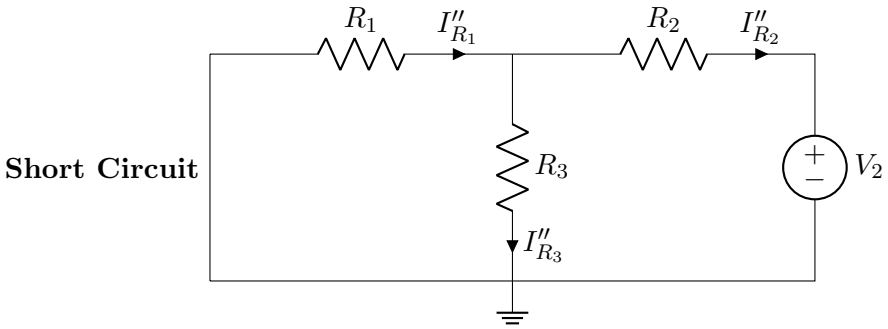


Figure 3: Modified circuit with V_2 active and V_1 replaced by a short circuit, giving the isolated contribution I''_{R_3}.

3. Switch off RPS2 and put a short-circuit wire in its place (Figure 2), leaving RPS1 on. Measure the isolated contribution I'_{R_3}.
4. Turn RPS2 back on and instead short RPS1 (Figure 3). Measure the isolated contribution I''_{R_3}.
5. Check that $I_{R_3} = I'_{R_3} + I''_{R_3}$ holds, then repeat Steps 2–4 for a second combination of source voltages.

Theoretical Calculations

With R_3 removed as the branch of interest and $R_1 = 325\,\Omega$, $R_2 = 550\,\Omega$, $R_3 = 960\,\Omega$ fixed for both trials:

Case 1: Only V_1 Active (V_2 Short-Circuited)

With V_2 shorted, R_2 and R_3 appear in parallel from node A to ground:

$$R_{23} = \frac{R_2 R_3}{R_2 + R_3} = \frac{550 \times 960}{550 + 960} = 349.67\,\Omega, \qquad R_{\text{eq1}} = R_1 + R_{23} = 325 + 349.67 = 674.67\,\Omega.$$

The total current leaving V_1 is $I'_{\text{total}} = V_1 / R_{\text{eq1}}$, giving node voltage $V'_A = I'_{\text{total}} \times R_{23}$, and hence

$$I'_{R_3} = \frac{V'_A}{R_3}.$$

Case 2: Only V_2 Active (V_1 Short-Circuited)

With V_1 shorted, R_1 and R_3 appear in parallel from node A to ground:

$$R_{13} = \frac{R_1 R_3}{R_1 + R_3} = \frac{325 \times 960}{325 + 960} = 242.80\,\Omega, \qquad R_{\text{eq2}} = R_2 + R_{13} = 550 + 242.80 = 792.80\,\Omega.$$

The total current leaving V_2 is $I''_{\text{total}} = V_2 / R_{\text{eq2}}$, giving node voltage $V''_A = I''_{\text{total}} \times R_{13}$, and hence

$$I''_{R_3} = \frac{V''_A}{R_3}.$$

Applied to Trial 1 ($V_1 = 10\,\text{V}$, $V_2 = 10\,\text{V}$)

$$I'_{R_3} = \frac{10/674.67 \times 349.67}{960} = 5.39\,\text{mA}, \qquad I''_{R_3} = \frac{10/792.80 \times 242.80}{960} = 3.19\,\text{mA},$$

$$I_{R_3} = I'_{R_3} + I''_{R_3} = 5.39 + 3.19 = 8.58\,\text{mA} \approx 8.59\,\text{mA}.$$

Applied to Trial 2 ($V_1 = 15\,\text{V}$, $V_2 = 10\,\text{V}$)

Since V_2 is unchanged, I''_{R_3} is the same as in Trial 1 (3.19 mA), while I'_{R_3} scales with V_1:

$$I'_{R_3} = \frac{15/674.67 \times 349.67}{960} = 8.09\,\text{mA},$$

$$I_{R_3} = I'_{R_3} + I''_{R_3} = 8.09 + 3.19 = 11.28\,\text{mA} \approx 11.29\,\text{mA}.$$

Both results match the calculated values reported in Table 2 and Table 3, confirming the theoretical circuit model.

Data Collection

Table 2: Experimental Data for the Verification of Superposition Theorem – Trial 1 ($V_1 = 10\,\text{V}$, $V_2 = 10\,\text{V}$ nominal)

Configuration	V_1 **(V)**	V_2 **(V)**	**Measured** I_{R_3} **(mA)**	**Calculated** I_{R_3} **(mA)**
Both Sources Active	10	10	8.61	8.59
Only V_1 Active	10	0	5.36	5.39
Only V_2 Active	0	10	3.21	3.19

Table 3: Experimental Data for the Verification of Superposition Theorem – Trial 2 ($V_1 = 15\,\text{V}$, $V_2 = 10\,\text{V}$ nominal)

Configuration	V_1 **(V)**	V_2 **(V)**	**Measured** I_{R_3} **(mA)**	**Calculated** I_{R_3} **(mA)**
Both Sources Active	15	10	11.3	11.29
Only V_1 Active	15	0	8.11	8.09
Only V_2 Active	0	10	3.2	3.19

Note: Data transcribed from the instructor-reviewed handwritten laboratory data sheet (Group 04).

Result Analysis

Adding up the two individually-measured branch currents from each trial gives a value that lines up well with what was measured when both sources were on at the same time:

- **Trial 1**: $I'_{R_3} + I''_{R_3} = 5.36\,\text{mA} + 3.21\,\text{mA} = 8.57\,\text{mA}$ (Measured total = 8.61 mA).
- **Trial 2**: $I'_{R_3} + I''_{R_3} = 8.11\,\text{mA} + 3.2\,\text{mA} = 11.31\,\text{mA}$ (Measured total = 11.3 mA).

To check this more rigorously, the measured "Both Active" current can be compared against the theoretical value using

$$\%\ \text{Error} = \left|\frac{\text{Theoretical} - \text{Measured}}{\text{Theoretical}}\right| \times 100\%,$$

which gives about 0.23% for Trial 1 ($|8.59 - 8.61|/8.59 \times 100\%$) and about 0.09% for Trial 2 ($|11.29 - 11.3|/11.29 \times 100\%$). Both numbers are small, so the Superposition Theorem holds up well in both trials.

Discussion and Sources of Error

None of the percentage errors we calculated went above 0.25%, so overall the results back up the theory very well. Still, a few things in the real setup keep it from matching the ideal calculations exactly:

1. **Resistor Tolerance:** R_1, R_2, and R_3 are not exactly their labeled values – every resistor is only guaranteed to within a manufacturer tolerance – so there's already a small gap between what the theory assumes and what's actually in the circuit.
2. **Power Supply Regulation:** The DC supply output can shift slightly once it's under load, and the short-circuit substitution used to deactivate a source is never perfectly ideal (a real wire still has some resistance), both of which add a bit of error into the trials.
3. **Multimeter Loading:** An ammeter isn't a true zero-resistance short – it has some small burden resistance – so just connecting the meter changes the circuit slightly from what's assumed in the calculations.

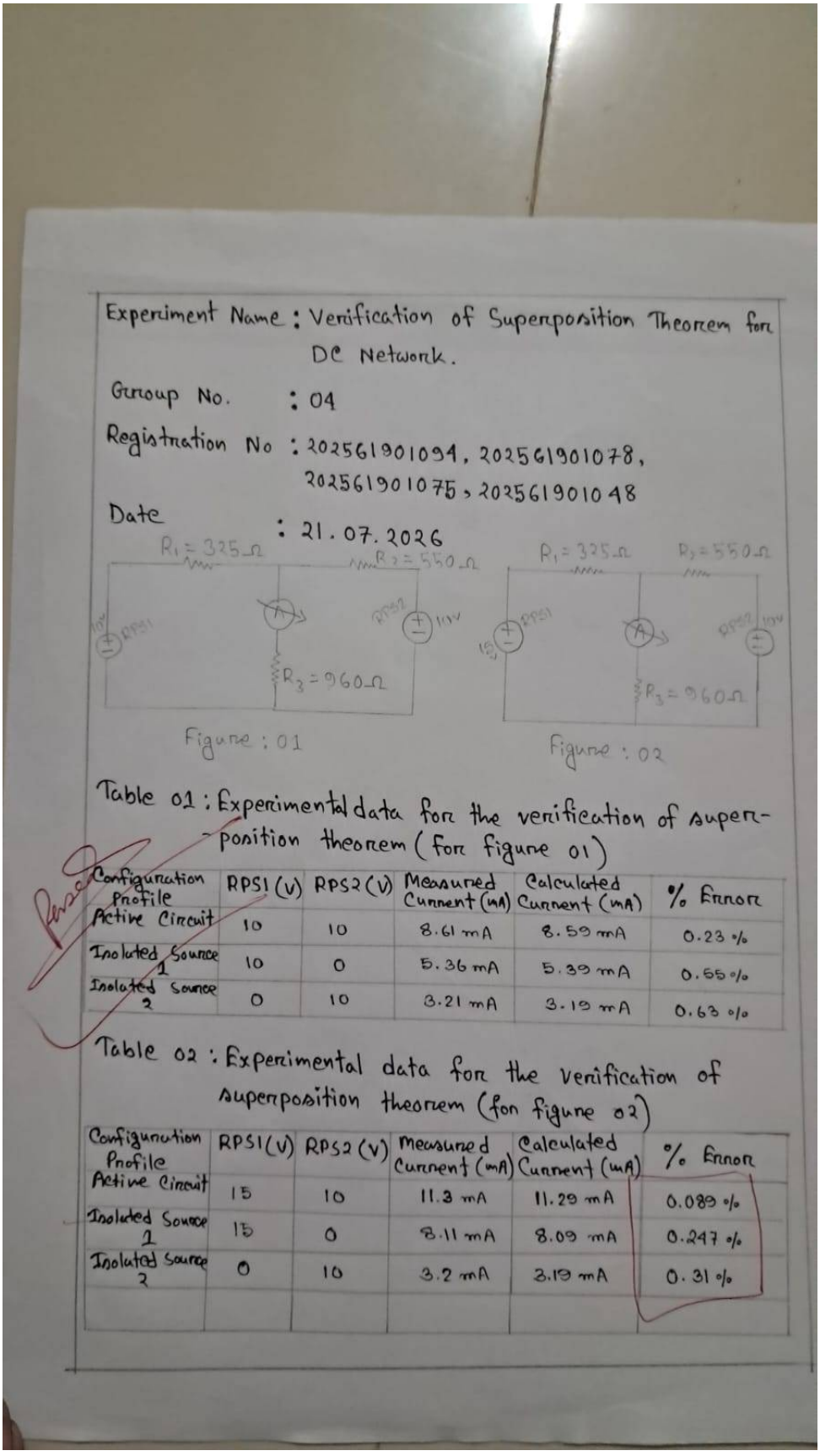
Conclusion

This experiment confirmed that the Superposition Theorem holds for a linear, multi-source DC network. Adding together the branch currents measured with each source acting alone gave a total that was very close to what was measured with both sources on together, in both of the two trials, with percentage errors under 0.25%. This confirms that superposition is a valid way to break a multi-source circuit down into simpler single-source problems, as long as you remember it only works for currents and voltages and not directly for power.

References

1. Alexander, C. K., & Sadiku, M. N. O. (2016). *Fundamentals of Electric Circuits* (6th ed.). McGraw-Hill Education. (Chapter 4: Circuit Theorems).
2. Hayt, W. H., Kemmerly, J. E., & Durbin, S. M. (2012). *Engineering Circuit Analysis* (8th ed.). McGraw-Hill Education. (Chapter 5: Useful Circuit Analysis Techniques).
3. Boylestad, R. L. (2015). *Introductory Circuit Analysis* (13th ed.). Pearson. (Chapter 9: Network Theorems).

Appendix: Signed Data Sheet



Experiment Name : Verification of Superposition Theorem for DC Network.

Group No. : 04

Registration No : 202561901094, 202561901078, 202561901075, 202561901048

Date : 21.07.2026

Figure : 01

Figure : 02

Table 01 : Experimental data for the verification of superposition theorem (for figure 01)

Configuration Profile	RPS1 (V)	RPS2 (V)	Measured Current (mA)	Calculated Current (mA)	% Error
Active Circuit	10	10	8.61 mA	8.59 mA	0.23 %
Isolated Source 1	10	0	5.36 mA	5.39 mA	0.55 %
Isolated Source 2	0	10	3.21 mA	3.19 mA	0.63 %

Table 02 : Experimental data for the verification of superposition theorem (for figure 02)

Configuration Profile	RPS1 (V)	RPS2 (V)	Measured Current (mA)	Calculated Current (mA)	% Error
Active Circuit	15	10	11.3 mA	11.29 mA	0.089 %
Isolated Source 1	15	0	8.11 mA	8.09 mA	0.247 %
Isolated Source 2	0	10	3.2 mA	3.19 mA	0.31 %

Figure 4: Instructor-signed laboratory data sheet for this experiment.